



NSW Education Standards Authority

2022 HIGHER SCHOOL CERTIFICATE EXAMINATION

Food Technology

**General
Instructions**

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black pen

**Total marks:
100****Section I – 20 marks** (pages 2–6)

- Attempt Questions 1–20
- Allow about 35 minutes for this section

Section II – 50 marks (pages 9–18)

- Attempt Questions 21–26
- Allow about 1 hour and 25 minutes for this section

Section III – 15 marks (page 33)

- Attempt Question 27
- Allow about 30 minutes for this section

Section IV – 15 marks (page 34)

- Attempt Question 28
- Allow about 30 minutes for this section

Section I

20 marks

Attempt Questions 1–20

Allow about 35 minutes for this section

Use the multiple-choice answer sheet for Questions 1–20.

- 1** An individual is diagnosed with anaemia.
- Which dietary supplement would best address this disorder?
- A. Vitamin C
 - B. Iron tablet
 - C. Dietary fibre
 - D. Sodium powder
- 2** What is the process of removing moisture from a food product?
- A. Drying
 - B. Freezing
 - C. Fermentation
 - D. Pasteurisation
- 3** A food product developer has completed the product specifications in the development of a new food product.
- What is the next step?
- A. Market research
 - B. Feasibility study
 - C. Development of a prototype
 - D. Production process development
- 4** For which of the following is seasonal employment an expected working condition?
- A. Fruit pickers
 - B. Forklift drivers
 - C. Horticulturalists
 - D. Production line engineers

- 5 Which food category is most likely to prevent the development of type 2 diabetes when regularly consumed?
- A. Fats and oils
 - B. Meat and dairy
 - C. Breads and cereals
 - D. Fruits and vegetables
- 6 What is the purpose of an emulsifier in food manufacturing?
- A. To prevent drying
 - B. To prevent oxidation
 - C. To combine oil and water
 - D. To extend the product's shelf life
- 7 During times of strong economic growth, many food manufacturers expand their product range.
- What is the best explanation for this?
- A. Market concerns
 - B. Consumer demands
 - C. Company profitability
 - D. Technological developments
- 8 Peeled, sliced and frozen potatoes that are sold in the food retail sector are an example of
- A. functional food.
 - B. value-added food.
 - C. low-cost nutritious food.
 - D. food to be consumed on the go.

- 9** Athletes consume food products with increased protein to enhance their performance.

These products are an example of

- A. supplements.
- B. enriched foods.
- C. functional foods.
- D. active non-nutrients.

- 10** A small sachet is inserted into a food package to remove gases during storage.

What type of packaging is this?

- A. Active
- B. Vacuum
- C. Sous vide
- D. Barrier specific

- 11** An apple producer has redesigned their fruit punnets to include a convenient carry handle.

Which of the following is most likely to have influenced this development?

- A. Market concerns
- B. Decreasing costs
- C. Technological developments
- D. Concerns for the environment

- 12** Producers can use technology to monitor irrigation remotely.

Which sector of the food industry is using this technology?

- A. Food retail
- B. Agriculture and fisheries
- C. Food service and catering
- D. Food processing and manufacturing

- 13** An individual is producing protein-destroying antibodies.

What is the individual most likely to be experiencing?

- A. Food poisoning
- B. Allergic reaction
- C. Diabetic episode
- D. Food intolerance

- 14** Which method of preservation relies on lactic acid to extend the shelf life of a food product?

- A. Chilling
- B. Dehydration
- C. Fermentation
- D. Freezing

- 15** A honey producer has applied to maintain the use of their product name based on flora that is unique to their geographic region.

What aspect of a SWOT analysis would have identified the need for this application?

- A. Strength
- B. Weakness
- C. Opportunity
- D. Threat

- 16** Two Australian dairy manufacturers have merged their operations and will make use of multiple processing plants across the country. As a result, their production capabilities will increase.

Which level of operation is represented in this merger?

- A. Household
- B. Multinational
- C. Small business
- D. Large company

- 17** The health of which specific group in Australia is most likely to be affected by geographic location, high mortality rates, predisposition to obesity, hypertension and cardiovascular disease?
- A. Athletes
 - B. Middle-aged men
 - C. Pregnant and lactating women
 - D. Aboriginal and Torres Strait Islander People

- 18** During a quality control inspection at a manufacturing facility, mould has been identified in a batch of freeze-dried pineapple.

What is the most likely cause of the growth of mould?

- A. Enzymatic activity
 - B. High moisture content
 - C. Inappropriate packaging
 - D. Incorrect storage conditions
- 19** A sustainable seafood partnership was developed in response to unethical aquaculture practices.

What is the primary external factor that has influenced the development of this partnership?

- A. The political environment
 - B. The economic environment
 - C. The ecological environment
 - D. The technological environment
- 20** A team of researchers investigates the use of electric motors for agricultural machinery in order to reduce carbon emissions associated with food production.

Which government body is most likely to be responsible for this research?

- A. CSIRO
- B. FSANZ
- C. NSW Food Authority
- D. Department of Agriculture, Water and the Environment

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Centre Number

Food Technology

Sections II, III and IV

Answer Booklet

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Student Number

Instructions

- Write your Centre Number and Student Number at the top of this page.
- Use this answer booklet for Sections II, III and IV.
- Follow the instructions inside this booklet as to where you should write your answers.

Please turn over

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Do NOT write in this area.

Section II

50 marks

Attempt Questions 21–26

Allow about 1 hour and 25 minutes for this section

Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.

Question 21 (4 marks)

- (a) Identify ONE function of packaging. **1**

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- (b) Recommend a suitable packaging material for refrigerated cookie dough. **3**

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Please turn over

Question 22 (12 marks)

An organisation is adopting a technology that can be used to rescue surplus food and distribute it via an online marketplace.

- (a) Outline the benefits of this technology for the environment. 3

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- (b) How does ONE piece of legislation affect the operation of a surplus food rescue organisation? 3

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Question 22 continues on page 13

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Question 22 (continued)

- (c) Explain storage and distribution considerations for rescued surplus food. Support your response with examples.

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End of Question 22

Please turn over

Question 23 (6 marks)

Justify the inclusion of TWO specific dietary supplements in contributing to a balanced diet for pregnant and lactating women.

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Question 24 (8 marks)

- (a) An organisation is repurposing food nearing the end of its shelf life into food products such as pesto, smoothies and pizzas. **3**

Why have these food products been developed?

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- (b) Analyse the impact of the micro-environment in the development of a successful food product. Support your response with an example. **5**

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Question 25 (12 marks)

- (a) An example of an adult's daily diet is provided below.

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Breakfast: bacon and fried egg on white bread roll with butter and BBQ sauce

Lunch: meat pie and sauce

Dinner: deep fried fish with chips

Snacks: potato chips and a 250 g bag of confectionery

Beverage: coffee, regularly consumed

Outline ONE dietary disorder that may develop as a result of this diet.

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- (b) Identify a different dietary disorder to the one outlined in part (a) and propose lifestyle practices that may improve an individual's nutritional status for this disorder.

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Question 25 continues on page 17

Question 25 (continued)

(c) How should a health food company advertise responsibly?

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End of Question 25

Please turn over

Question 26 (8 marks)

- (a) How can a principle of preservation assist in ensuring that dairy products are safe for consumption?

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- (b) Assess the suitability of ONE preservation process for extending the shelf life of a meat product. Provide examples to support your response.

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Section III — Answer Question 27 on pages 19–25

27

Question Number

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If you require more space to answer Section III, you may ask for an extra writing booklet.

If you have used an extra writing booklet for Section III, tick here.

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Section IV — Answer Question 28 on pages 26–32

28

Question Number

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If you require more space to answer Section IV, you may ask for an extra writing booklet.

If you have used an extra writing booklet for Section IV, tick here.

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Food Technology

Section III

15 marks

Attempt Question 27

Allow about 30 minutes for this section

Answer the question on pages 19–25 of the Sections II, III and IV Answer Booklet. Extra writing booklets are available.

Question 27 (15 marks)

A food retailer has opened a new store that provides a range of self-serve food options. This includes locally sourced and culturally considered ingredients to meet the consumer demands of the surrounding area.

- (a) Identify TWO career opportunities this store has created in the food retail sector. **2**
- (b) Explain quality management considerations that need to be in place to ensure the self-serve food options are safe for public consumption. **5**
- (c) Evaluate the use of TWO promotional strategies suitable for a locally sourced food product sold by the food retailer. Provide examples to support your response. **8**

Please turn over

Section IV

15 marks

Attempt Question 28

Allow about 30 minutes for this section

Answer the question on pages 26–32 of the Sections II, III and IV Answer Booklet. Extra writing booklets are available.

Your answer will be assessed on how well you:

- demonstrate knowledge and understanding relevant to the question
 - apply course concepts to food technology issues
 - communicate ideas and information using appropriate terminology and relevant examples
 - present a logical and cohesive response
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Question 28 (15 marks)

Assess the roles of government organisations, food manufacturers and community groups in improving the diet and health of Australians.

End of paper